Failure-Aware and Confidence-Calibrated Adaptive Retrieval-Augmented Generation System

23CSE399 Project Phase 1

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Agenda

1. Problem Statement
2. Research Gaps
3. Literature Survey
4. Objectives
5. Architecture
6. Functionalities
7. Software Requirements
8. Project Timeline
9. Conclusion
10. Future Scopes
11. References

Introduction

- RAG retrieves documents using embedding similarity and lexical matching between query and documents
- Modern systems rank results using similarity scores, lexical matching (e.g., BM25), and hybrid fusion methods
- These ranking signals measure relevance, not reliability or correctness
- Weak, inconsistent, or partially irrelevant documents can still be retrieved
- Existing signals act as proxy indicators, not a true retrieval reliability score
- Advanced systems introduce re-ranking and validation steps, but reliable failure detection remains unsolved

Research Gaps

- No unified retrieval reliability score
- Lack of calibrated confidence scores
- No robust pre-generation failure detection
- No modeling of retrieval stability
- No unified multi-signal failure framework

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
1	H. Soudani, E. Kanoulas, and F. Hasibi	Why Uncertainty Estimation Methods Fall Short in Retrieval-Augmented Generation: An Axiomatic Analysis, 2025	Defines axioms for reliable uncertainty estimation in RAG. Shows token probability and entropy methods ignore retrieval quality and grounding. Emphasizes need for retrieval-aware uncertainty estimation. Evaluation Metrics Used: Expected Calibration Error (ECE), Brier Score, Negative Log Likelihood (NLL), and AUROC.	Does not propose a practical mechanism to compute retrieval reliability using measurable signals such as similarity score consistency, retrieval variance, or agreement.
2	L. Ngartera, S. Nadarajah, and R. Koina	Bayesian RAG: Uncertainty-Aware Retrieval for Reliable Financial Question Answering, 2026	Uses Bayesian embeddings to model retrieval uncertainty and improve reliability. Evaluation Metrics Used: Accuracy, F1 Score, Expected Calibration Error (ECE), Brier Score, Negative Log Likelihood (NLL), and AUROC.	Does not explicitly detect retrieval failures using similarity consistency or agreement signals.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
3	Michael Oehri, Jannis Vamvas, et al.	Trusted Uncertainty in Large Language Models: A Unified Framework for Confidence Calibration and Risk-Controlled Refusal, 2025	Proposes unified confidence framework using likelihood, entropy, and self-consistency signals. Enables calibrated confidence estimation and risk-controlled refusal in LLM responses. Evaluation Metrics Used: Expected Calibration Error (ECE), Brier Score, Negative Log Likelihood (NLL)	Does not incorporate retrieval reliability estimation or similarity-based retrieval failure signals in RAG systems.
4	Y. Gao, Y. Xiong, X. Gao, et al.	Retrieval-Augmented Generation for Large Language Models: A Survey, 2023	Provides comprehensive overview of RAG architecture, retrieval methods, and reliability challenges. Identifies uncertainty estimation and retrieval reliability as key limitations. Evaluation Metrics Used: Exact Match (EM), F1 Score, Accuracy, BLEU, ROUGE	Does not propose practical methods to quantify retrieval reliability using similarity consistency or agreement signals.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
5	S. Es, J. James, L. Espinosa-Anke, and S. Schockaert	RAGAS: Automated Evaluation of Retrieval-Augmented Generation, 2023	Introduces metrics such as context relevance, faithfulness, and grounding to evaluate RAG performance. Enables automated evaluation without human annotation. Evaluation Metrics Used: Context Precision, Context Recall, Faithfulness	Provides evaluation metrics but does not detect retrieval failures or estimate retrieval reliability dynamically during inference.
6	V. Karpukhin et al.	Dense Passage Retrieval for Open-Domain Question Answering, 2020	Introduces dense retrieval using neural encoders and embedding similarity scoring. Improves retrieval accuracy over traditional keyword-based methods. Evaluation Metrics Used: Recall-k (e.g., Recall@20, Recall@100), Top-k Retrieval Accuracy, Exact Match (EM), and F1 Score.	Uses similarity scores for document ranking but does not analyze similarity patterns to estimate retrieval reliability or detect retrieval failures.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
7	S.-Q. Yan, J.-C. Gu, Y. Zhu, and Z.-H. Ling	Corrective Retrieval-Augmented Generation, 2024	Introduces a retrieval evaluator to assess relevance and correctness of retrieved documents. Improves retrieval quality by correcting weak or irrelevant retrieval results. Evaluation Metrics Used: Exact Match (EM), F1 Score, Accuracy, Recall-k, Expected Calibration Error (ECE)	Requires a separate 'evaluator' model to trigger web searches, adding architectural complexity. It corrects retrieval but doesn't quantify a confidence score for the user.
8	P. Lewis et al.	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks, 2020	Introduces RAG framework combining dense retrieval and generation using embedding similarity. Improves factual accuracy by grounding responses in retrieved knowledge. Evaluation Metrics Used: Exact Match (EM), F1 Score, Accuracy, Perplexity	Does not estimate retrieval reliability or detect weak retrieval before generation.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
9	A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi	Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection, 2023	Introduces reflection tokens to evaluate relevance and usefulness of retrieved documents. Enables the model to critique retrieval quality and improve response grounding. Evaluation Metrics Used: Exact Match (EM), F1 Score, Accuracy, Human Evaluation	Relies on 'reflection tokens' and fine-tuning, which increases computational cost and latency. It does not assess the retriever's raw stability before generation begins.
10	O. Khattab and M. Zaharia	ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT, 2020	Introduces token-level late interaction similarity scoring using contextual embeddings. Improves retrieval precision and semantic matching between query and documents. Evaluation Metrics Used: Mean Reciprocal Rank (MRR), Recall-k, Normalized Discounted Cumulative Gain	Does not compute retrieval reliability or detect weak retrieval using similarity consistency signals.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
11	J. Song, X. Wang, J. Zhu, et al.	RAG-HAT: A Hallucination-Aware Tuning Pipeline for LLM in Retrieval-Augmented Generation, 2024	Introduces hallucination-aware tuning pipeline using retrieval-guided training signals. Improves factual grounding and reliability of generated responses in RAG systems.	Does not explicitly estimate retrieval reliability or detect weak retrieval using similarity consistency signals.
12	H. Ding, L. Pang, Z. Wei, H. Shen, and X. Cheng	Rowen: Adaptive Retrieval-Augmented Generation for Hallucination Mitigation in LLMs, 2025	Introduces adaptive retrieval mechanism to dynamically improve document choice. Enhances retrieval relevance and reduces hallucinations in generated responses.	Does not compute retrieval reliability scores using similarity consistency or retrieval agreement signals.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
13	Z. Sun, X. Zang, K. Zheng, et al.	ReDeEP: Detecting Hallucination in Retrieval-Augmented Generation via Mechanistic Interpretability, 2025	Uses mechanistic interpretability to analyze internal model activations during generation. Enables detection of hallucinations caused by weak or incorrect grounding. Evaluation Metrics Used: Hallucination Detection Accuracy, Precision, Recall	Focuses on generation analysis but does not estimate retrieval reliability using similarity consistency or agreement signals.

Literature Survey

S.No	Authors	Paper Title (Year)	Inference from Paper	Open Problem (For Proposed Work)
1	S. Robertson, H. Zaragoza	The Probabilistic Relevance Framework: BM25 and Beyond (2009)	Introduces probabilistic ranking using BM25. Ranks documents based on probability of relevance. Strong baseline for IR systems.	Relies on term-frequency matching; fails in semantic understanding and contextual retrieval.
2	V. Karpukhin et al.	Dense Passage Retrieval for Open-Domain QA (2020)	Uses dual-encoder dense embeddings for retrieval. Outperforms BM25 significantly in QA tasks.	Requires large labeled data; struggles with domain-specific retrieval and hallucination control.
3	N. Reimers, I. Gurevych	Sentence-BERT (2019)	Introduces siamese BERT to generate sentence embeddings. Enables fast semantic similarity search.	Does not directly optimize for retrieval grounding or factual consistency in RAG pipelines.

Literature Survey

S.No	Authors	Paper Title (Year)	Inference from Paper	Open Problem (For Proposed Work)
4	N. Pipitone et al.	LegalBench-RAG (2024)	Benchmark for evaluating retrieval quality in legal RAG systems. Focuses on precise snippet-level retrieval.	Lacks integration with generation reliability; retrieval still evaluated separately from hallucination control.
5	I. Chalkidis et al.	LexGLUE: Legal NLP Benchmark (2022)	Provides multi-task benchmark for legal language understanding. Helps evaluate model generalization.	Focuses on NLU tasks, not retrieval + generation pipelines (RAG).
6	D. Hendrycks et al.	CUAD: Contract Understanding Dataset (2021)	Large expert-annotated dataset for legal contract analysis. Enables clause extraction and classification.	Dataset-heavy approach; lacks retrieval + reasoning integration and scalability for real-time systems.

Literature Survey

S.No	Authors	Paper Title (Year)	Inference from Paper	Open Problem (For Proposed Work)
7	T. Zhang et al.	RAFT: Adapting LLM to Domain-Specific RAG (2024)	Proposes Retrieval-Augmented Fine-Tuning. Trains model to ignore irrelevant documents and improve reasoning.	Still dependent on retrieval quality; does not explicitly measure retrieval reliability or consistency.
8	M. Douze et al.	FAISS Library (2025)	Efficient vector similarity search for large-scale embeddings. Backbone for dense retrieval systems.	Focuses only on indexing/search; no semantic reasoning or hallucination mitigation.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
9	Huaiyu Qin, Chunyu Wei, et al.	COUNTERFACTUAL REASONING FOR RETRIEVALAUGMENTED GENERATION, 2026	Tackles misleading but correlated retrieval results. Uses counterfactual reasoning to identify causally relevant evidence	Computationally heavy due to multiple counterfactual queries.
10	Ori Yoran, Tomer Wolfson, et al	MAKING RETRIEVAL-AUGMENTED LANGUAGE MODELS ROBUST TO IRRELEVANT CONTEXT, 2024	Focuses on filtering irrelevant retrieved documents. Uses NLI-based relevance checking	Only **filters** does not correct or replace bad retrieval. Performance depends on NLI model

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
11	Zhengbao Jiang, Frank F. Xu, et al.	Active Retrieval Augmented Generation, 2023	iterative retrieval during generation instead of one-shot retrieval. Performs well on multi hop QA.	High latency + compute cost. Assumes retrieved docs are reliable.
12	Timo Schick, Jane Dwivedi-Yu, et al	Toolformer: Language Models Can Teach Themselves to Use Tools, 2024	Trains LLMs to call external tools. Learns tool usage via self-supervision.	Requires pretraining modifications. Does not explicitly detect or fix bad retrieval quality

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
13	Patrick Lewis et al.	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks	Combines retrieval + generation to improve factual accuracy and reduce hallucination.	Depends on retrieval quality; slow and lacks multi-step reasoning.
14	Lun-Chi Chen, Mayuresh Sunil Pardeshi , Yi-Xiang Liao, Kai-Chih Pai	Application of retrieval-augmented generation for interactive industrial knowledge management via a large language model	BM25 + embeddings improves retrieval performance for complex queries.	No reranking or self-correction; weak generation stage.

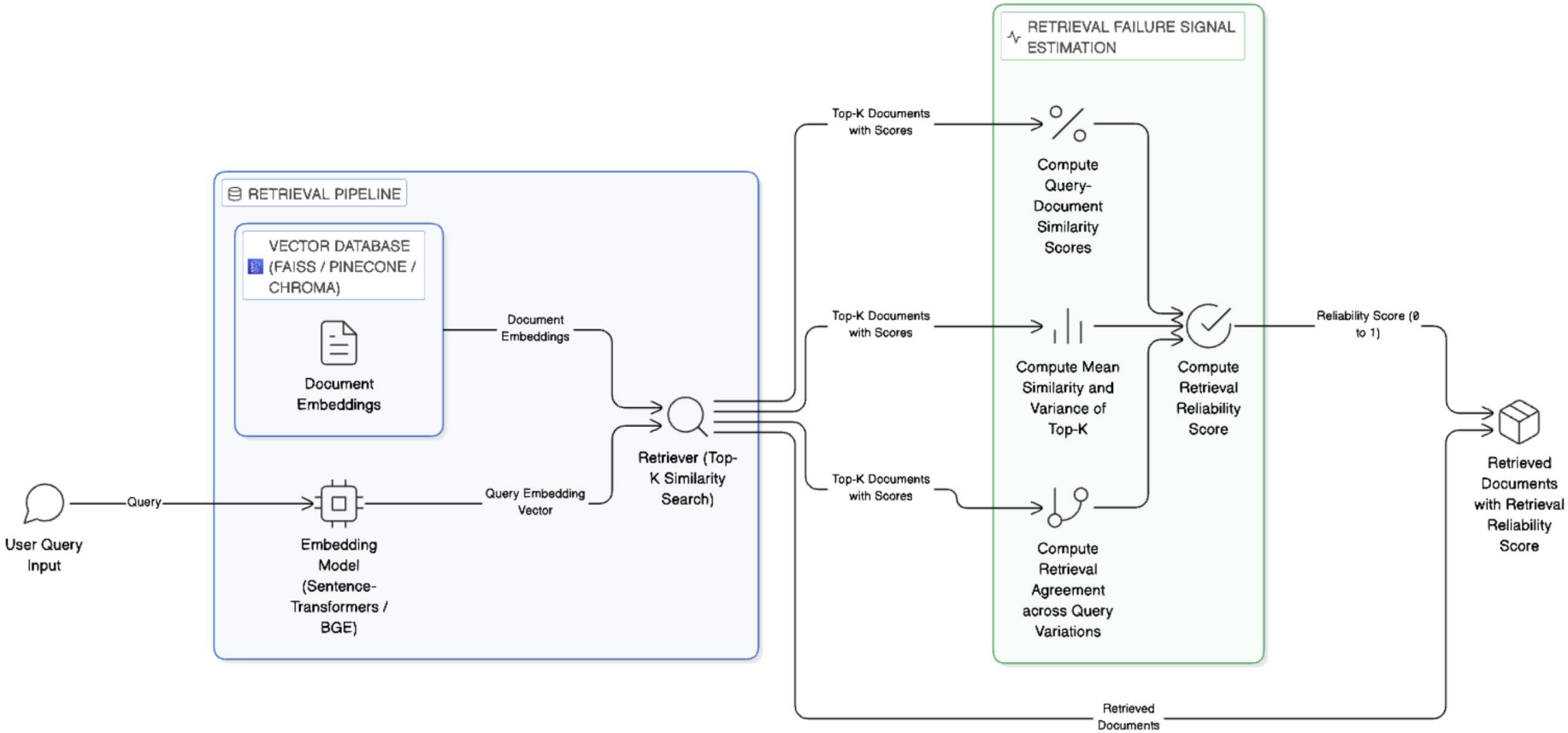
Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
15	J. Rayo, R. de la Rosa, and M. Garrido	A hybrid approach to information retrieval and answer generation for regulatory texts	Practical RAG system with hybrid retrieval and reranker improves real-world performance.	Static knowledge base; no hallucination control or learning mechanism.
16	Shan Xu, Zhaokun Yan, Chengxiao Dai and Fan Wu1	MEGA-RAG: Multi-Evidence Guided RAG	Uses multiple answers + verification to reduce hallucination and improve reliability.	Computationally expensive and slow for real-time use.

Objectives

- Current RAG models rank using similarity scores but cannot explicitly detect retrieval failure
- Similarity scores do not guarantee correctness or consistency
- Advanced models (e.g., Self-RAG, CRAG) assess retrieval quality but do not compute explicit reliability from measurable signals
- Poor or inconsistent retrieval can still pass through and affect generation
- A mechanism is needed to estimate retrieval failure using signals like similarity, top-k consistency, and agreement
- This work proposes a method to compute a retrieval failure signal for detecting unreliable retrieval

Architecture of the Proposed System



Dataset Description

SciFact is a scientific fact verification dataset.

Task:
Given a scientific claim, classify it as:
- SUPPORT
- REFUTE
- NEUTRAL

based on evidence from research papers.

{ "id": 42,
"claim": "Vitamin D supplementation reduces the risk of respiratory infections.",
"label": "SUPPORT",
"evidence": [{ "doc_id": "DOC_1023", "sentences": [1],
"text": "Vitamin D supplementation was found to reduce the incidence of acute respiratory infections in clinical trials." }]
}

- Requires reasoning over scientific text
- Not keyword matching → needs semantic retrieval
- Contains real-world claims from research
- Includes supporting and contradicting evidence

{ "id": 123,
"claim": "Text of the claim",
"label": "SUPPORT / REFUTE / NEUTRAL",
"evidence": [{ "doc_id": "document_id", "sentences": [0, 2],
"text": "Evidence sentence(s)" }] }

Basic RAG

Query → Embedding → Dense Retrieval (FAISS) → Top-K Docs → LLM (per-doc) → Weighted Voting → Final Answer

Embedding:
- Model: Specter / MiniLM
- Converts claim → dense vector

Retriever:
- FAISS (IndexFlatIP)
- Cosine similarity (L2 normalized)
- Top-K = 3

Generator:
- Qwen / Mistral (7B)
- Prompt-constrained classification

Multi-doc reasoning:
- Each document processed independently

Aggregation:
- Convert similarity → probability
- Weighted voting across docs

Final prediction:
- Label with highest total probability

Functionalities of the Proposed System

Core RAG Pipeline

- Convert input query into dense embedding
- Retrieve top-k documents using cosine similarity
- Compute and rank query–document similarity scores
- Provide retrieved context for generation

Retrieval Failure Score Module

- Compute similarity magnitude of top-k documents
- Analyse top-k similarity consistency
- Measure retrieval agreement among documents
- Aggregate signals into a retrieval reliability score
- Detect weak or unreliable retrieval before generation

Software/Tools Requirements

Programming Language:

- Python

Frameworks:

- HuggingFace Transformers
- Sentence-Transformers
- LangChain / LlamaIndex
- FAISS / ChromaDB

Models:

- Embedding Model: BGE-large or MiniLM , GloVe ,Word2Vec
- Generator Model: Mistral-7B (future integration) , Phi-3 , Phi-2 , Llama-3,Qwen

Platform:

- Kaggle GPU Environment

BM25-Based RAG Model

BM25-Based RAG for Scientific Claim Verification

- Dataset: SciFact
- Goal: Classify claims as SUPPORT / REFUTE / NEUTRAL

Methodology:

- Claim
- BM25 Retrieval (find relevant scientific papers)
- Top Documents (context)
- LLM (FLAN-T5)
- Prediction (SUPPORT / REFUTE / NEUTRAL)

Larger models like
Qwen2.5-7B-Instruct and
Mistral-7B-Instruct-v0.1
were attempted but not used due to hardware constraints

BM25-Based RAG Model

Performance

- Accuracy: 0.35
- F1 Score: 0.17

Observations

- Model predicts NEUTRAL frequently
- Indicates: Weak retrieval, poor context, uncertain predictions

Limitations

- BM25 lacks semantic understanding
- Retrieved documents may be irrelevant
- No retrieval validation or correction

Corrective RAG Model

CRAG-Based RAG for Scientific Claim Verification

- Dataset: SciFact
- Goal: Classify claims as SUPPORT / REFUTE / NEUTRAL

Methodology:

- Claim
- Initial Retrieval (BM25 / Retriever)
- Retrieve top-k scientific documents
- Retrieved Documents Evaluation (CRAG Evaluator)
- Assess relevance & quality of retrieved documents
- Assign confidence: **Correct / Incorrect / Ambiguous**

•**Corrective Actions:**

•**Correct:** Refine documents (extract key evidence)

•**Incorrect:** Discard and perform **web search / external retrieval**

•**Ambiguous:** Combine both refined + external knowledge

Corrective RAG Model

Model Choices:
LLM used Qwen 2.5B
Results:
Initial accuracy = 0.25
Final accuracy = 0.60
Accuracy gain = 0.35
Average faithfulnes = 0.65

Observations:
Initially when the retrieval quality was bad the model used to misclassify and the maximum achievable accuracy when given poor retrievals was 0.25 but when identified and corrected retrieval qulality the accuracy gain was 0.35 and the final accuracy achieved was 0.60

RAFT – based learning model

Dataset: SciFact
Goal: Classify claims as SUPPORT / REFUTE / NEUTRAL

Methodology:
• Dataset Input (in .json format)
•Initial Retrieval (BM25 / Dense Retriever)
•Top-k Retrieved Documents (Relevant + Distractors)
•RAFT Fine-Tuned LLM
•Chain-of-Thought Reasoning (Grounded in Evidence)
•Prediction (SUPPORT / REFUTE / NEUTRAL)
Fine-Tuning Strategy:
- Uses Parameter-Efficient Fine-Tuning (PEFT)
- Common approach: LoRA (Low-Rank Adaptation)

Trainable Parameters:
- Total Model Parameters: ~7 Billion
- Trainable Parameters: 2M
  Percentage Updated: 0.03%

RAFT – based learning model

Performance Metrics

- Accuracy: 0.56
- Macro-F1: 0.51
- Precision / Recall: 0.53 / 0.50

Class-wise Performance

- SUPPORT F1: 0.57
- REFUTE F1: 0.46
- NEUTRAL F1: 0.56

Grounding & Retrieval

- Faithfulness: 0.74
- Improved retrieval diversity
- Better alignment with evidence

Limitations:

- Limited ability to learn new reasoning patterns
- Cannot fully handle contradictions
- Performance still depends on retrieval quality

Comparative Analysis

Model	Accuracy	F1 / Macro-F1	Faithfulness	Hallucination	Recall@k	Key Strengths	Key Limitations
BM25	0.35	0.17	0.418	0.581	0.88	Simple lexical retrieval	No semantic understanding, irrelevant docs, no validation
Basic RAG	0.72	—	—	0.48	0.9	High recall (0.9), good baseline performance	High hallucination, no retrieval validation
RAFT	0.56	0.51	0.74	—	0.91	Better grounding, improved retrieval diversity	Cannot handle contradictions, depends on retrieval quality
CRAG	0.60	0.62	0.65	0.35	0.78	Corrects poor retrieval, +0.35 accuracy gain	No explicit reliability score, no failure detection

Results

BM25 MODEL

```
df.to_csv("bm25_scifact_final_results.csv", index=False)

Loading corpus...
Docs: 5183
Loading claims...
Claims: 100
Loading model...
Loading weights: 100%|▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰| 282/282 [00:01<00:00, 238.62it/s, Materializing param=shared.weight]
The tied weights mapping and config for this model specifies to tie shared.wei

Running...
0 → Pred: NEUTRAL | GT: NEUTRAL
10 → Pred: NEUTRAL | GT: SUPPORT
20 → Pred: NEUTRAL | GT: SUPPORT
30 → Pred: NEUTRAL | GT: SUPPORT
40 → Pred: NEUTRAL | GT: REFUTE
50 → Pred: NEUTRAL | GT: SUPPORT
60 → Pred: NEUTRAL | GT: NEUTRAL
70 → Pred: NEUTRAL | GT: NEUTRAL
80 → Pred: NEUTRAL | GT: NEUTRAL
90 → Pred: NEUTRAL | GT: SUPPORT

FINAL RESULTS:
Accuracy: 0.35
F1 Score: 0.1728395061728395
Recall@k: 0.88
Faithfulness: 0.41878709277336584
Hallucination: 0.5812129072266342
```

bm25_scifact_final_results.csv

1 to 10 of 100 entries Filter

pred	gt	recall	faith	halluci
NEUTRAL	NEUTRAL	1	0.0	1.0
NEUTRAL	SUPPORT	1	0.2631577562327599	0.7368422
NEUTRAL	SUPPORT	1	0.4285708163274053	0.5714291
NEUTRAL	NEUTRAL	1	0.3999996000004	0.6000003
NEUTRAL	NEUTRAL	1	0.22222197530891633	0.7777780
NEUTRAL	REFUTE	1	0.3749997656251465	0.6250002
NEUTRAL	REFUTE	1	0.30769207100609924	0.6923079
NEUTRAL	SUPPORT	1	0.4999991666680556	0.5000008
NEUTRAL	SUPPORT	1	0.571427755103207	0.4285722
NEUTRAL	REFUTE	1	0.44444395061783265	0.5555560

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Timeline

- Abstract Submission • 5/2/2026
- Literature Survey • 17/2/2026
- System Architecture Design Finalization • 5/3/2026
- Embedding Model Integration and Vector Database Setup • 20/3/2026
- Basic Retrieval Pipeline Implementation • 5/4/2026
- Implementation of Failure Aware Signals • 30/4/2026

Conclusion

This work proposes a failure-aware RAG framework to improve system reliability.

This phase focuses on detecting retrieval failures using similarity score, top-k consistency, and retrieval agreement signals.

The system estimates retrieval reliability and enables early detection of weak retrieval.

This forms the foundation for future confidence estimation and adaptive RAG systems.

Future Scopes

Enterprise Knowledge Management involves retrieving and utilizing information distributed across multiple heterogeneous sources such as documents, reports, and internal knowledge bases.

In real-world organizations:

- Information is fragmented across formats (PDFs, text, databases)
- The same concept may appear in multiple sources with variations
- Manual search is inefficient and error-prone

Goal:
To build a system that does not just generate answers, but generates answers supported by verifiable sources.

Datasets

Dataset	Data Type / Format	Simulated Enterprise Source
BEIR Benchmark	Text (multi-domain: QA, news, biomedical)	Mixed internal knowledge sources
MS MARCO	Text (web passages, semi-structured)	Internal documents / reports
SciFact	Text (claims + evidence abstracts)	Research reports / knowledge base
Natural Questions	Text (Wikipedia-based QA)	User search queries
TriviaQA	Text (multi-source QA)	Knowledge base queries
LoTTE	Text (forum/discussion threads)	Slack / Jira / internal chats
BEIR Subsets (FiQA, TREC-NEWS, NFCorpus)	Text (domain-specific corpora)	Finance / news / domain data
CodeSearchNet	Code + Docstrings + Comments	GitHub / internal codebase
Stack Overflow	Text (Q&A, informal discussions)	Developer tickets / discussions
DocVQA	PDF / Document Images + Text	Enterprise documents (PDFs)
SlideVQA	Images (presentation slides)	PPTs / architecture diagrams
MMQA	Text + Tables + Images	Mixed enterprise data
MP-DocVQA	Multi-page documents (PDF + images)	Long reports / manuals

Future Scopes (Use case)

Product & Engineering Docs

Engineers waste 30-40% of their time searching across Confluence, GitHub, Jira, Figma exports, and PDF specs trying to answer questions like:

- *"What's the API rate limit for the payments service?"*
- *"Which team owns the auth microservice?"*
- *"What does this architecture diagram mean?"*
- *"Was this bug fixed in the last sprint?"*

Your multimodal heterogeneous RAG answers all of these from one interface.

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श्रद्धावान् लभते ज्ञानम्

THANK YOU